CHAPTER – 1

Classification of Recyclable Waste Using Deep Learning

INTRODUCTION

1.1 Overview

Waste management involves managing a variety of waste materials like gases, liquids or solids. These actions include transporting, gathering, knowing handling and recycling waste. Agriculture, industry, societies and other organization produce waste substances. Constantly changing waste properties in the process is significant. Thus, they must be considered in sampling. Measures to manage and control the production, storage, collection, transport, recovery and disposal of waste are taken using appropriate technology and scientific powers.

In order to stay sustainable, human societies must process an enormous amount of waste. Getting rid of waste is a huge issue as they keep producing waste regularly and some are tough to treat. Nuclear waste, for example, has specific treatment issues. In the same fashion, waste that is extremely poisonous also has its own treatment challenges. The volume and the characteristics both play a critical role in waste processing.

Recycling Waste Is Important.

The volume of waste substances is increasing at a very high rate due to major driving forces like urbanization, industrialization and increase in population. If this happens, it can have damaging impacts as the waste gets released in huge volumes into the air, water and soil. Various sources produce waste that is harmful to the quality of life.

The annual statistic is more than 5.4 million tons of recyclable waste is sent sent to landfill or incinerated. Besides, the typical American produces about 4.4 pounds of garbage a day. Individuals from all walks of life generate recycling waste every day. So, the goal of the project “Web Based Waste Classification System” is to automate waste classification and stop recyclable waste going to a landfill.

Waste sorting at home, in the office and almost everywhere by different people is now common. Nonetheless, due to the various types and varieties of waste we generate, the classification of wastes is still a challenging task for the labourers. Therefore, developing a Classifier System can bring workers to a better solution for accurate waste classification.

The user can upload an image of waste as input to the web-based waste classification system project. After that, the system predicts the class of waste whether it is recyclable or bio-degradable using the model. The project employs top-notch image classification models which include CNN, VGG16, ResNet50, MobileNet and EfficientNet for extraction of features and classification.

We are designing a web-based system to classify wastes using deep learning. It assists with.

1.2 Problem Statement

A traditional method of disposal is the manual segregation of waste. However, it relies on human observation who may have trouble identifying it as many of its looks similar

1.2.1 Manual Classification Issues

- Needs the help of humans.Human errors are common
- Takes a long time.
- Managing waste on a large scale is not easy.

1.2.2 Similar Appearance of Waste Materials

Various waste materials show similarity:

- Color
- Shape
- Texture
- Outer Looks

This causes commotion in manual classification.

1.2.3 Lack of Automated Systems

After a thorough paper review conducted on the used solid waste, it was established that the geographic location, occupation and habit of the resident plays a key role in shaping solid waste.

1.2.4 Dataset Challenges

- Labeled datasets are limited
- Datasets are not balanced
- Image quality is different
- Light and background is different

Aim of the Project

The project aims to design a system that completely automates the segregating of waste using deep learning advancement. Specifically, the system identifies and categorizes waste.

1.3 Objectives

1. To design and implement automated waste classification system.
2. To apply online image preprocessing techniques.
3. Deep learning models will also extract richer features from this for further analysis.
4. To improve classification accuracy using transfer learning.
5. To compare multiple deep learning architectures.
6. To balance the dataset for better performance.
7. To reduce dependency on manual identification.
8. To spread awareness about proper waste segregation.
9. To develop a scalable and user-friendly system.

1.4 Applications

The proposed system has various applications in the real world:

- **Smart Waste Management**

Automatic segregation of waste must be done at homes, offices, schools, public places.

- **Recycling Industry**

The management of a scrap is a major issue in a manufacturing unit.

- **Municipal Waste Management**

These records aid local municipal authorities in enhancing waste collection and segregation systems in the region.

- **Environmental Protection**

Minimizes pollution and encourages proper waste disposal practices.

- **Educational Purpose**

Management and recycling can be taught in schools and colleges to make them aware of it.

- **Smart City Applications**

Smart waste management systems can be installed in smart cities.

- **Public Awareness Systems**

Provides disposal instructions and encourages people to follow proper waste segregation methods.

1.5 Scope of the Project

The goal of this project is to develop an intelligent system which is capable of detecting waste of different categories.

Future enhancements include:

- Real-time waste detection
- Multi-waste object detection
- Increasing the range of waste categories.
- Improved Dataset and Accuracy
- Environmental Monitoring and Analytics

Chapter-2

LITERATURE SURVEY

2.1 Introduction

The document discusses the waste classification system's research work done by various researchers. The waste classification system helps to sort various kinds of waste materials at the initial stage of waste management. Increasing interest for segregation of waste is being automated to enhance efficiency of waste classification for recycling. For waste segregation in the real world, scientists and engineers use machine learning and deep learning techniques. The technique of identifying object/scene activity in a digital image or video is called an image recognition system.

2.2 Research Paper 1

Title

Organic and Recyclable Waste Classification using Deep Learning Methods

Method Used

- VGG16
- VGG19
- ResNet18
- DenseNet121
- DenseNet169
- Simple CNN
- Transfer Learning

Working

- Images of waste are collected from Kaggle dataset.
- The image gets preprocessed and resized.
- Deep learning models are used to extract image features automatically.
- Models categorize waste as organic and recyclable.
- The best performing model is selected based on performance selection.

Accuracy

Using customized VGG16 with transfer learning 95.59%.

Advantages

• High classification accuracy.

• Performance is improved by transfer learning.

• Extraction of Features Automatically.

• Allows for several deep learning structures.

Limitations

- Needs a large training set.
- Accuracy affected by image quality.
- High computational requirements.

1.3 Research Paper 2 –

Title

Classification of Recyclable Waste using Deep Learning Approaches

Method Used

• ResNet-50

• Custom 5-Layer CNN

• Image Classification Techniques

• Deep Learning Frameworks

Accuracy

A custom CNN model achieves 91.84%

Advantages

- It is usable feature for web and mobile application integration.
- Reduced training data requirements
- Quicker training processes

Limitations

- Requires proper hyperparameter tuning
- Performance depends on dataset quality
- Deep models require more memory

Feature	Paper 1	Paper 2	Paper 3
Paper Title	Organic and recyclable waste classification using deep learning methods	Classification of recyclable waste using deep learning architectures	Recyclable Waste Classification Using Computer Vision And Deep Learning
Dataset Used	Combined Kaggle dataset with 27,603 color images, split into Organic and Recyclable classes.	Dataset from Kaggle containing 22,500 color images (227X227 pixels) of Organic and Recyclable waste.	TrashNet dataset featuring 2,527 images across 6 classes (paper, glass, plastic, metal, cardboard, trash).
Different Models Tested	VGG16, VGG19, ResNet18, DenseNet121, DenseNet169, and a Simple CNN.	ResNet-50 and a custom-built 5-layer CNN architecture.	12 CNN variants (including Inception V3, Xception, VGG16, VGG19, ResNet variants, MobileNet, DenseNet201, NASNet) paired with 3 classifiers (SVM, Sigmoid, SoftMax).
Best Model & Accuracy	Customized VGG16 (with transfer learning) achieved 95.59% test accuracy.	Custom CNN achieved 91.84% accuracy, outperforming the ResNet-50 (81.34%).	VGG19 + SoftMax classifier achieved the best result with 87.9% test accuracy.
Limitations	Highly dependent on hardware resources (requires a capable GPU). Accuracy relies heavily on the diversity of training data, and class imbalances can cause model bias.	The research focuses strictly on binary classification (Organic vs. Recyclable), which may limit granular sorting.	Failed to classify food waste due to dataset constraints (reduced to 5 classes). The model also struggles with images that do not have the dataset's standard white background.

TABLE 2.1 COMPARISION OF RESEARCH PAPER

Chapter-4

SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10/11 or Linux
- **Programming Language:** Python 3.14
- **Development Platform:** JupyterNotebook / VS Code / Google Colab
- **Frameworks:** TensorFlow.
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Flask
- **Dataset Source:** Kaggle and TrashNet datasets
- **Hardware Requirement:** Minimum 8 GB RAM, Intel i5 Processor or above

Chapter-5

IMPLEMENTATION DETAILS

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5.1 Step 1: System Design Selection

The model proposed here is based on Deep Learning architectures for garbage classification. We created a simple convolutional neural network (CNN) model first due to the reason that these models are able to automatically learn important image features like shape, colour, texture, and even patterns of objects from waste images.

Following that, sophisticated architectures, such as VGG16, ResNet50, MobileNet, and EfficientNet, were examined on the input data to achieve enhanced classification accuracy and improved feature extraction capabilities. Due to better performance and higher accuracy all these versions of CNN are widely used in image classification tasks.

The proposed model will be a web-application under which a user may upload the image of the waste. Additionally, the trained model predicts if the waste is recyclable or biodegradable..

5.2 Step 2: Dataset Used

We gathered images of waste from publicly available data sets and online sources for this project. More specifically, we used kaggle and trashNet.

Images of different waste belonging to the following classes are available in the data.

- Waste that is recyclable
- (3 words): Biodegradable waste.

Turning and Rotating.

Zoom Change.

5.5 Step 5: Image Reshaping

All images are resized to a fixed dimension for compatibility with deep learning models. The input images are converted to tensor format of dimension size compatible for CNN and Vision Transformer models training and testing.

5.6 Step 6: Label Encoding

Each waste category is assigned a numerical label.

For example:

- Recyclable → 0
- Biodegradable → 1

The labels are converted into **one-hot encoded vectors** so that the deep learning models can correctly classify multiple categories during supervised learning.

5.7 Step 7: Data Splitting

The learning model parameters are defined using the training dataset and the evaluation of the performance of learned model on unseen Waste images is done by testing dataset.

5.8 Step 8: Model Training

Various Deep learning models such as:

- **CNN**
- **VGG16**
- **ResNet50**
- **MobileNet**
- **EfficientNet**

were trained using the prepared dataset.

During training:

- Forward propagation computes predictions
- Backpropagation updates weights
- Adam optimizer minimizes loss
- Categorical Crossentropy is used as the loss function

Transfer learning with pre-trained ImageNet weights was also applied to improve feature extraction and reduce training time.

Training was performed for multiple epochs until the model achieved stable accuracy and reduced loss.

5.9 Step 9: Dataset Balancing and Optimization

Initially, the dataset was imbalanced, which caused poor accuracy and biased predictions.

To solve this issue:

- Data augmentation techniques were applied
- Smaller classes were increased
- Duplicate and unnecessary images were removed
- Class balancing techniques were used

After balancing the dataset:

- Model performance improved
- Overfitting was reduced
- Classification became more accurate

5.10 Step 10: Model Testing

Various performance metrics were used to evaluate the trained models:

- Accuracy
- Loss
- Validation Accuracy
- Training Accuracy
- Precision
- Recall

It was observed that there was evaluation comparative of performance of different model. Among all the models, transfer learning models like MobileNet outperformed basic CNN models in terms of a score. The selected model is able to achieve a good classification accuracy and performance is achieved.

5.11 Step 11: Output Generation

The final trained system accepts a waste image as input and predicts the most probable waste category as output.

The system automatically performs:

- Image preprocessing
- Feature extraction
- Classification

before displaying the result.

The output includes:

- Predicted waste category

This helps users understand how to dispose of waste correctly and promotes better waste management practices.

Chapter-6

RESULTS

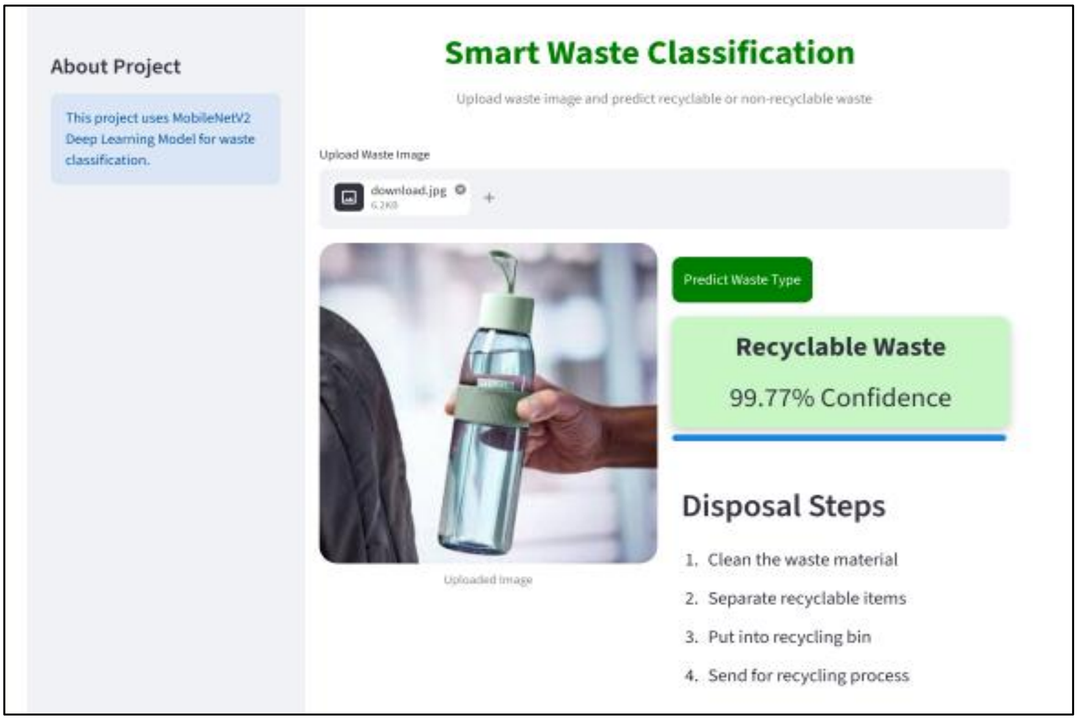

Figure 6.1: Web interface showing recyclable waste prediction with confidence score and disposal guidance.

	precision	recall	f1-score	support
0	0.96	0.89	0.93	2513
1	0.88	0.95	0.91	1999
accuracy			0.92	4512
macro avg	0.92	0.92	0.92	4512
weighted avg	0.92	0.92	0.92	4512

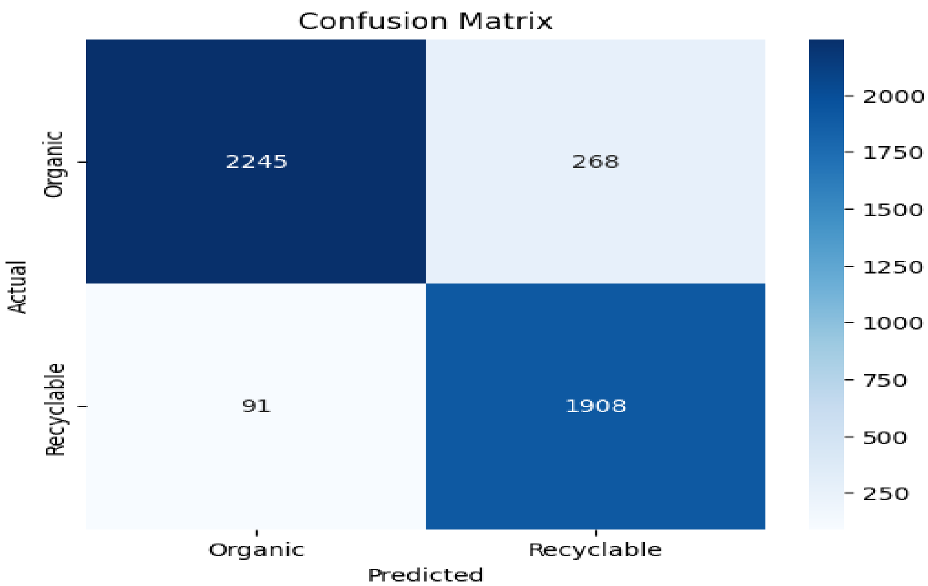

Figure 6.2: Web interface displaying non-recyclable waste classification with disposal instructions.

Figure 6.3: Confusion matrix showing the accuracy of the waste classification model.

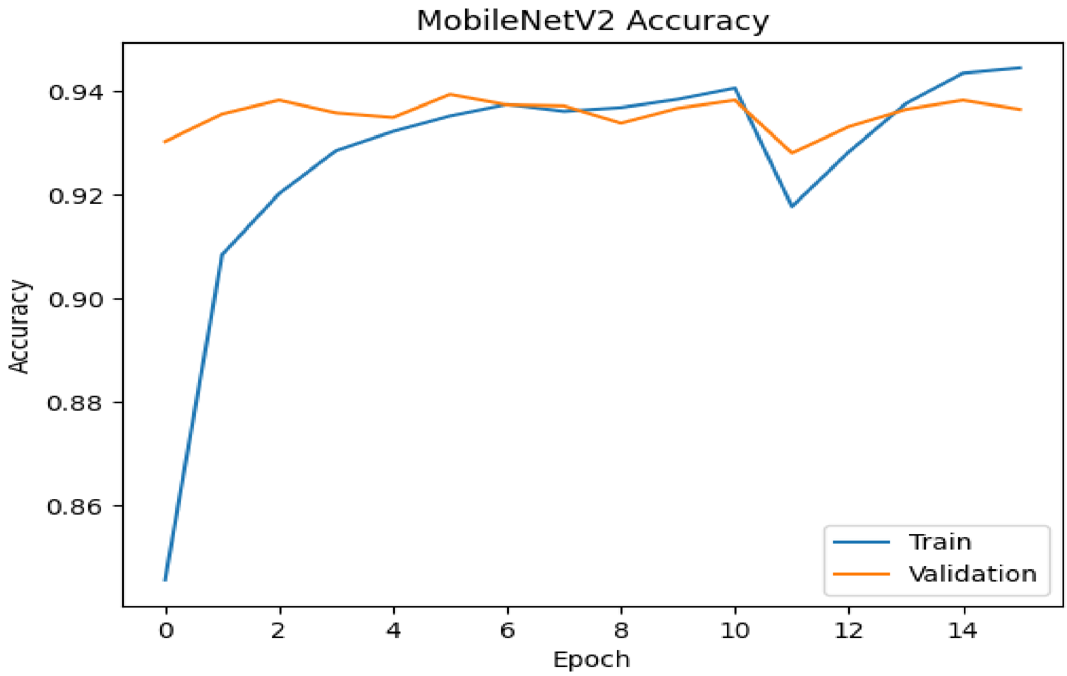

Figure 6.4: MobileNetV2 Accuracy VS Epoch Graph

CONCLUSION

The project entitled "Web Based Waste Classification System Using Deep Learning" shows the use of Artificial Intelligence and Deep Learning in the Waste Management. The system automatically classifies waste images as recyclable or biodegradable. The project employs state-of-the-art image classification techniques and deep learning architectures. The initial phase of testing shall involve the use of multiple deep learning models CNN, VGG16, ResNet50, MobileNet, EfficientNet to classify waste. If we develop a dataset which is free from above issues, it could be more effective at solving waste recognition and classification problem.

In order to decrease the issues with these data, a few preprocessing techniques are applied followed by data augmentation. To enhance overall classification accuracy, further balancing of the dataset was performed. The project also focuses on preperation of an appropriate dataset, image prepossessing, data augmentation, transfer learning, balancing the dataset to reduce training-testing bias. It was also shown how models could be optimized.

The waste classification is done with very less human effort. It will help in building better smart waste management solutions. Users will also get live hands-on experience of deep learning, computer vision, transfer learning and image classification techniques.

The system has significant potential for implementation in the real-world. It can be employed in the intelligent waste management framework, recycling sector, municipal waste management, smart city missions, environmental awareness initiatives and more. Also, It can be used for education and research purpose.